



Fig. 13: Chaff shell firing (left) and animated evading (right)

During Doolittle Raid, epitomised by Hollywood movie Pearl Harbour, carriers and associated ships sneaked towards Japan under total radio silence. They had to do that to prevent the Japanese intercepting their communication and hence detecting their sneaking. In today's scenario, radio silence will require all radio communications and radar emissions switched off, which is known as emission control (EMCON) state

to operate in a certain frequency band to exploit the property of those frequencies. Say, for example, a radar that requires projecting a highly narrow beam with greater resolution requires an X-band frequency. So if the ESM module picks up a particular radar frequency, the type of radar operating could easily be found out. The apparent strength of the signal indicates proximity of that radar.

Assume that the frequency used in the radar seeker of a cruise missile is found out. It is also found out that its strength is gradually increasing. In this case, we can easily say that a cruise missile is on its way for a kill.

'Know thy enemy' is a rule in war. So every country tries to know all possible electronic signatures (nature of signals) of its enemy's radar systems. Signals Intelligence (SIGINT) is a broad name for such operations. Nations do not hesitate to mount specific 007 kind of operations. All ESM systems maintain a library of known electronic signatures. Just by comparing the incoming signature with the library, the ESM module can easily find out the kind of system that is operating.

ESM is not only limited to radars

but is used for communications also. Enemy communications will be intercepted by consistently sweep tuning to the entire frequency band. A radio voice communication detected indicates that there is enemy radio activity. The intelligence personnel embedded with the carrier strike group (CSG) decode and translate enemy communications.

The enemy also has ESM facility. So if the CSG has to sneak near enemy shores, it has to maintain radio silence. In that case, the enemy ESM facilities serve the CSG to operate, like a blind man with acute ears.

Electronic counter measures.

Denying the enemy the advantage of electromagnetic waves is called an ECM. This includes two approaches, namely, hard kill or jamming, and soft kill or spoofing.

Hard kill/jamming. Jamming is synonymous to shouting loudly so that others cannot converse. If noise is injected in the telephone line, it naturally prevents a speaker from hearing what the other person is saying. Similarly, when a particular frequency has to be jammed, the ECM module generates and broadcasts a powerful signal in that fre-

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- Voltage Upto 600V AC/DC
- Resistance Upto 20MΩ
- Jaw Opening Upto 25mm
- Auto Power Off
- Non Contact Voltage
- Data Hold & Max Function

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